Topic 16: Tapeout

ECE 9433: Agile System-on-Chip Design

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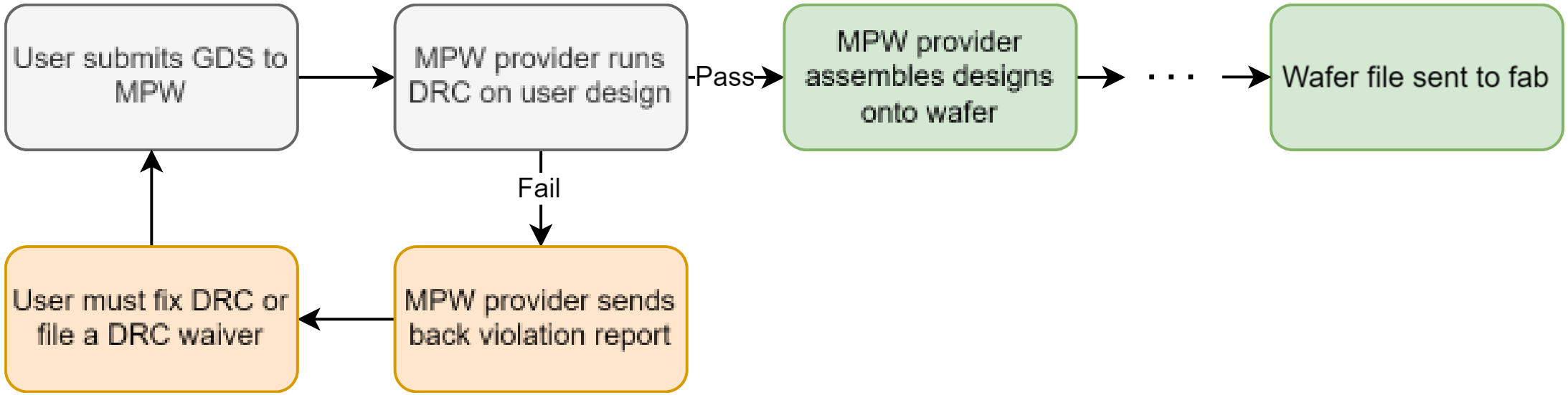
What is a “Tapeout”?

- Send chip layout file (GDS/OASIS) to be manufactured
- Origin
  - Chip design files predated the internet (or were at least too large to send over the internet / ARPAnet)
  - Magnetic tapes containing the chip design were physically sent (mail/courier) to the foundry
- Usage
  - *Noun*: **tapeout** or **tape-out**. Ex: “I have a tapeout in two weeks”
  - *Verb*: **tape out**. Ex: “I need to tape out in two weeks”

Tapeout Process

- Only companies doing large volume productions work directly with the fab
- Most users will work with a multi-project wafer (MPW) provider
- A designer will submit their design to the MPW provider, then the MPW provider will integrate all designs onto a single wafer and send out the wafer design file to the fab for production

Tapeout Process



DRC Violations

- If DRC passes for you, but the MPW provider reports violations, you need to check your DRC settings
- Check with *all* DRC rule decks (base, metal, antenna, etc.)
- Turned on all necessary settings in the DRC deck
- Many DRC rule files have switches for various settings, e.g.
  - Check for seal ring / don’t check for seal ring
  - Check only block-level rules / check for chip-level rules
  - Selection for the correct metal stack
  - Several more…

DRC Waivers

- Why do design rules exist?
  1) To ensure the proper manufacture of the chip
  2) To prevent damage / defects to other chips
  3) To prevent damage to the fab equipment
- If a rule is only for (1), there is a chance the rule can be bent
- Many rules are binary passing/failing when in reality, they are on a spectrum
- You can file a DRC waiver if you believe you have a good reason to waive the rule

DRC Waivers

- Good reason: You need a metal line which is 2% longer than the max allowed length in order to form an antenna for your RF circuit
- Bad reason: You can’t get the tool to fix a metal spacing violation
- Bad reasons will get rejected
- Different foundries have different reputations
- Ex: TSMC is historically extremely strict. Violation = no tapeout
- Ex: Intel is historically a bit flexible, mainly because design & fab teams work together

Tapeout Timeline

- MPW providers reserve some (not much) time for iterating on DRC
- Could be anywhere from 3 days to a couple weeks
- If you miss a deadline, you can get dropped from the MPW
- You may have to pay a penalty and then wait for the next spot, which could be 6+ months away
- Once MPW is sent to fab, can expect diced chips back in ~3-6 months
  - Depends on several factors, e.g. technology node, how high is demand, etc.

Acknowledgement